## Project Details

### CLIENT INFORMATION

| | |
|---|---|
| Client Name | Charles |
| Contact Email | — |
| Contact Phone | — |

### INSTALLER / COMPANY

| | |
|---|---|
| Company | Afrocarb |
| Email | emmanuelnwanja@gmail.com |

### PROJECT LOCATION

| | |
|---|---|
| Address | — |
| City / State | Lekki, Lagos |
| Coordinates | 6.5432°N, 3.3675°E |

### PROJECT SPECIFICATIONS

| | |
|---|---|
| System Type | Grid-Tied + Battery |
| PV Capacity | 6.00 kWp |
| Battery | 5.00 kWh |
| PV Technology | TOPCon |

Installation Date: 15 April 2026 Generated: 21 April 2026 Design Lifetime: 25 Years

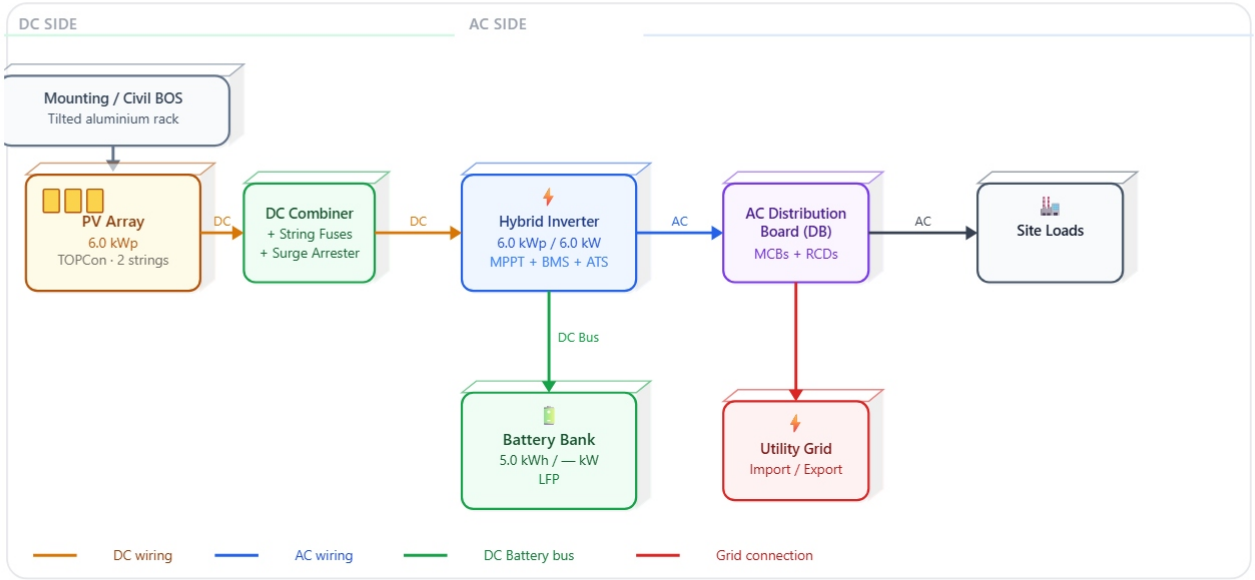
## System Schematic

Generated single-line system representation

## System Schematic

Grid-Tied + Battery · Rooftop (Tilted Rack)

6.0 kWp PV | 5.0 kWh BESS | ~2 strings | Tilt rack aligned to azimuth target | ~15 panels est.




| Component | Specification | Note |
|---|---|---|
| PV Array | 6.0 kWp · TOPCon | ~15 × 400 Wp panels est. · 2 strings · Tilt 5° · Az 180° |
| DC Combiner | String fuse box + surge arrester | One fuse per string. 1000 V DC rated (NEC/IEC 62548) |
| Mounting & Civil BOS | Tilted aluminium rack | Improves airflow and module temperature. |
| Hybrid Inverter | 6.0 kW AC output · — kW battery port | MPPT + built-in BMS interface · Grid-tied with storage |
| Battery Bank | 5.0 kWh · — kW · LFP | Min SOC —% · RTE —% · Strategy: self consumption |
| AC Distribution Board | MCBs + RCDs + SPD | Per load circuit. Earth leakage protection. IEC 60364 compliant |
| Grid Connection | Bidirectional meter | NERC net-metering compliant where available |

⚠ This schematic is a design-intent diagram generated from simulation parameters. Panel count (est. ~15) assumes 400 Wp average and installation assumptions for "Rooftop (Tilted Rack)". Actual string configuration, cable sizing, mounting checks, and protection device ratings must be verified by a qualified electrical engineer before installation. Compliant with NEC 2023, IEC 62548, and NERC grid connection guidelines.

| PV CAPACITY | BATTERY CAPACITY | ANNUAL GENERATION | YEAR 1 SAVINGS | PAYBACK PERIOD |
|---|---|---|---|---|
| 6.0 kWp | 5.0 kWh | 7,913 kWh | ₦293,769 | 25.0 yrs |

| Solar Fraction | Self-Consumption | NPV | IRR | LCOE | ROI |
|---|---|---|---|---|---|
| 66.2% | 1.3% | ₦-53,659,149 | 1,000.0% | ₦882.47/kWh | -129.0% |

## Uncertainty & Traceability

| P50 (MWh) | P90 (MWh) | P95 (MWh) | Engine Version |
|---|---|---|---|
| — | — | — | — |

Input Hash: —
Formula Hash: —
Weather Hash: —

## Units & Precision

All KPI values are rendered with explicit units and deterministic rounding.

| Storage precision | Display precision | Rounding mode |
| --- | --- | --- |
| Canonical numeric precision in compute payloads | Report-specific rounding for human readability | Default: half_away_from_zero |

## Formula References

Formula registry entries are scaffolded in backend and will appear here when V2 compute payload is enabled.

## Uncertainty Decomposition

| Weather | Model | Operations | Combined |
| --- | --- | --- | --- |
| 0.00% | 0.00% | 0.00% | 0.00% |

## Assumptions & Limitations

### Assumptions

- Weather input is based on available climatology/source datasets.
- Model coefficients follow currently active engine/formula versions.
- Financial escalations are applied as configured in design inputs.

### Limitations

- Outputs are deterministic given identical inputs, not guarantees of field outcomes.
- Data quality and missing-site context can materially affect model confidence.
- Results should be reviewed by qualified engineers before procurement/final sign-off.

## Provenance Audit

| | |
|---|---|
| Engine Version<br>– | Input Snapshot Hash<br>– |
| Formula Bundle Hash<br>– | Weather Dataset Hash<br>– |

## PV System Configuration

Solar array specifications and design parameters

| | | | |
|---|---|---|---|
| Technology<br>TOPCon | Array Tilt<br>5° | Array Azimuth<br>180° | System Losses<br>11% |
| Annual Degradation<br>0.4% | Analysis Period<br>25 years | Discount Rate<br>10% | Tariff Escalation<br>8%/yr |

## Battery Energy Storage System

Energy storage specifications and performance

| | | | |
|---|---|---|---|
| Rated Capacity<br>5 kWh | Power Rating<br>— | Chemistry<br>lfp | Round-Trip Efficiency<br>90% |
| Depth of Discharge<br>90% | Dispatch Strategy<br>Self Consumption | Annual Throughput<br>638 kWh | Annual Cycles<br>142 |

## Grid Tariff Plan

Electricity rate structure and utility details

| Tariff Name | Utility | Currency | Grid Topology |
|---|---|---|---|
| Nigeria MYTO Band C | All DisCos (12-16hr supply) | NGN | Grid-Tied + Battery |

Rate Schedule

| Season | Period | Rate/kWh |
|---|---|---|
| year_round | flat | ₦50 |

## Cost Analysis

Financial comparison and economic metrics

| Baseline Annual Cost | With Solar Cost | Annual Savings | LCOE |
|---|---|---|---|
| ₦22,038,082 | ₦21,744,313 | ₦293,769<br>1% reduction | ₦882.47/kWh |

| Peak Demand (Before Solar) | Peak Demand (After Solar) |
|---|---|
| 5,203.0 kW | 5,201.2 kW |

## Energy Source Comparison

Annual comparison with alternative energy sources

| Solar System | Grid Only | Diesel Generator | Petrol Generator |
|---|---|---|---|
| ₦21,744,313 | ₦22,038,082 | ₦141,507,025 | ₦111,963,143 |
| Your choice | Save ₦293,769/yr | Save ₦119,762,712/yr | Save ₦90,218,830/yr |

### Lifetime Cost (25 Years)

| | |
|---|---|
| Best Grid Only | ₦1,611,114,695 |
| Solar System | ₦1,639,638,411 |
| Petrol Gen. | ₦11,011,245,830 |
| Diesel Gen. | ₦13,916,799,784 |

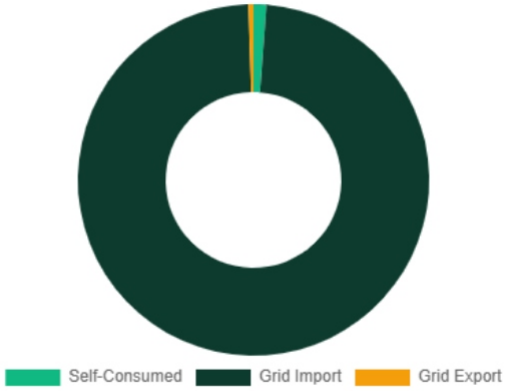
### Fuel Consumption Comparison (Annual)

| Energy Source | Fuel/yr | $CO_2$/yr (kg) | Cost/kWh |
|---|---|---|---|
| Solar | — | 186,886 | ₦882.47 |
| Grid Only | — | 189,412 | — |
| Diesel Gen. | 125,855 L | 337,293 | ₦321.25 |
| Petrol Gen. | 157,319 L | 363,408 | ₦254.18 |

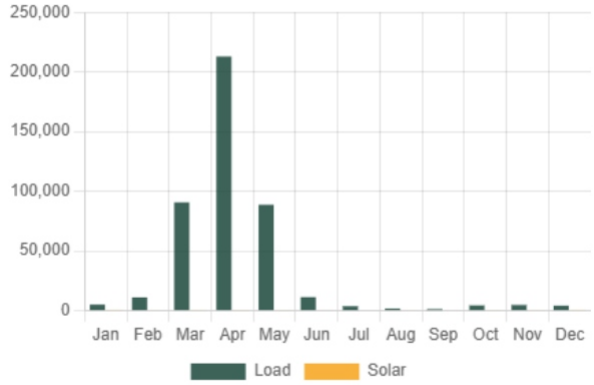
### Environmental Impact (Annual)

| 2,526 | 63.2 | 115 | 125,855 |
|---|---|---|---|
| kg $CO_2$ avoided vs grid | tonnes $CO_2$ avoided (lifetime) | trees equivalent per year | litres diesel avoided/yr |

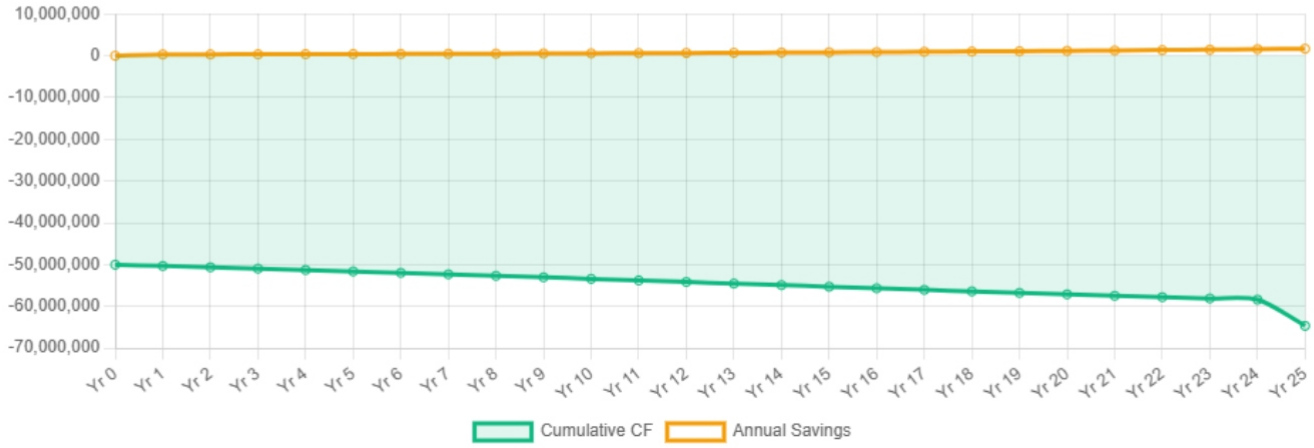
## Energy Split (Annual)




## Monthly Energy (kWh)




## Financial Projection




## Yearly Cashflow Projection

Annual financial breakdown over project lifetime

| Year | Savings | O&M | Net Cashflow | Cumulative | Generation |
|---|---|---|---|---|---|
| 0 | ₦0 | ₦0 | ₦-50,000,000 | **₦-50,000,000** | 0 kWh |

| | | | | | |
|---|---|---|---|---|---|
| 1 | ₦293,769 | ₦600,000 | ₦-306,231 | **₦-50,306,231** | 7,913 kWh |
| 2 | ₦315,843 | ₦630,000 | ₦-314,157 | **₦-50,620,387** | 7,878 kWh |
| 3 | ₦339,576 | ₦661,500 | ₦-321,924 | **₦-50,942,312** | 7,842 kWh |
| 4 | ₦365,091 | ₦694,575 | ₦-329,484 | **₦-51,271,796** | 7,807 kWh |
| 5 | ₦392,524 | ₦729,304 | ₦-336,779 | **₦-51,608,575** | 7,772 kWh |
| 6 | ₦422,019 | ₦765,769 | ₦-343,750 | **₦-51,952,325** | 7,737 kWh |
| 7 | ₦453,729 | ₦804,057 | ₦-350,328 | **₦-52,302,654** | 7,702 kWh |
| 8 | ₦487,822 | ₦844,260 | ₦-356,438 | **₦-52,659,092** | 7,667 kWh |
| 9 | ₦524,477 | ₦886,473 | ₦-361,996 | **₦-53,021,088** | 7,633 kWh |
| 10 | ₦563,886 | ₦930,797 | ₦-366,910 | **₦-53,387,998** | 7,599 kWh |
| 11 | ₦606,257 | ₦977,337 | ₦-371,080 | **₦-53,759,078** | 7,564 kWh |
| 12 | ₦651,811 | ₦1,026,204 | ₦-374,393 | **₦-54,133,471** | 7,530 kWh |
| 13 | ₦700,788 | ₦1,077,514 | ₦-376,726 | **₦-54,510,196** | 7,496 kWh |
| 14 | ₦753,445 | ₦1,131,389 | ₦-377,944 | **₦-54,888,141** | 7,463 kWh |
| 15 | ₦810,059 | ₦1,187,959 | ₦-377,900 | **₦-55,266,040** | 7,429 kWh |
| 16 | ₦870,927 | ₦1,247,357 | ₦-376,430 | **₦-55,642,470** | 7,396 kWh |
| 17 | ₦936,368 | ₦1,309,725 | ₦-373,356 | **₦-56,015,827** | 7,362 kWh |
| 18 | ₦1,006,727 | ₦1,375,211 | ₦-368,484 | **₦-56,384,310** | 7,329 kWh |
| 19 | ₦1,082,373 | ₦1,443,972 | ₦-361,599 | **₦-56,745,909** | 7,296 kWh |
| 20 | ₦1,163,702 | ₦1,516,170 | ₦-352,468 | **₦-57,098,377** | 7,263 kWh |
| 21 | ₦1,251,143 | ₦1,591,979 | ₦-340,836 | **₦-57,439,213** | 7,231 kWh |
| 22 | ₦1,345,154 | ₦1,671,578 | ₦-326,424 | **₦-57,765,637** | 7,198 kWh |
| 23 | ₦1,446,228 | ₦1,755,156 | ₦-308,928 | **₦-58,074,565** | 7,166 kWh |
| 24 | ₦1,554,898 | ₦1,842,914 | ₦-288,016 | **₦-58,362,581** | 7,134 kWh |

| | | | | | |
|---|---|---|---|---|---|
| 25 | ₦1,671,733 | ₦1,935,060 | ₦-6,263,327 | **₦-64,625,908** | 7,101 kWh |

## Environmental Impact Summary

Carbon offset and sustainability metrics

| Annual CO2 Avoided | Lifetime CO2 Avoided | Trees Equivalent | Diesel Avoided |
|---|---|---|---|
| 2,526 kg vs grid | 63.2 tonnes | 115 per year | 125,855 L per year |

## Design Warnings & Notes

Load profile has a very low load factor (1%) — peak demand (5203 kW) is 103× the average (50.28 kW). Ensure inverter and battery power ratings can handle peak transients.

## Monthly Energy Summary

Detailed monthly performance breakdown

| Month | Load | Solar Gen | Self-Used | Grid Import | Grid Export | Savings |
|---|---|---|---|---|---|---|
| Jan | 5,088 | 733 | 495 | 4,524 | 169 | ₦28,248 |
| Feb | 11,107 | 635 | 445 | 10,628 | 155 | ₦23,951 |
| Mar | 90,770 | 708 | 500 | 90,251 | 184 | ₦25,973 |
| Apr | 212,966 | 708 | 493 | 212,451 | 196 | ₦25,787 |
| May | 88,746 | 682 | 516 | 88,211 | 147 | ₦26,736 |
| Jun | 11,330 | 548 | 358 | 10,938 | 151 | ₦19,600 |
| Jul | 3,806 | 571 | 369 | 3,381 | 149 | ₦21,248 |
| Aug | 1,767 | 638 | 378 | 1,286 | 154 | ₦24,053 |
| Sep | 1,403 | 601 | 322 | 995 | 186 | ₦20,373 |
| Oct | 4,456 | 666 | 445 | 3,945 | 157 | ₦25,550 |
| Nov | 4,786 | 693 | 467 | 4,261 | 165 | ₦26,272 |
| Dec | 4,271 | 731 | 450 | 3,752 | 209 | ₦25,978 |
| **Total** | **440,494** | **7,913** | **5,237** | **434,619** | **2,021** | **₦293,769** |

### Disclaimer

This report is generated by the SolNuv platform for preliminary design and feasibility assessment purposes. Actual system performance may vary based on local conditions, equipment specifications, and installation quality. A detailed engineering study by a qualified professional is recommended.

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